

Makis Lam

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EDUCATION

University of Waterloo

Bachelor of Applied Science, Honours Mechanical Engineering, Co-op (3.81/4.0 GPA)

Sept. 2022 - Apr. 2027

Waterloo, ON

EXPERIENCE

FuturHand Robotics

Robotics Engineer - Dexterous Manipulation

May 2026 - Present

Pittsburgh, PA

- Researched multi-modal learning approaches to enable remote proprioceptive sensing, contact prediction, and contact force estimation for dexterous manipulation, deploying the inference pipeline onto a robotic hand at 90 Hz, leading to a first-author IEEE Conference for Robotics and Automation (ICRA) 2027 submission with CMU.
- Engineered 5+ tendon-driven, soft, dexterous robotic hands in Onshape, leveraging motion capture data for optimized joint kinematics and applying DFAM principles for flexible FDM printing, enabling factory pilot trials.
- Architecting an automated CAD pipeline via the Onshape API to generate monolithic robotic hands from optimized kinematics, accelerating design iterations by a targeted 10X to enable custom, task-driven designs.
- Integrated hardware and software architectures, programming ROS2 control nodes and designing custom PCBs.
- Conducted DFMEA and reliability testing to triage hardware issues and prioritize continuous design upgrades.
- Managed prototype-to-production vendor sourcing and trial negotiations to ensure compliance with design specs.
- Directed Onshape CAD migration; managed platform evaluation, pricing negotiation, and design standards.

Formlabs

Mechanical Design Engineer - Fuse X1 SLS R&D

Jan. - Apr. 2026

Somerville, MA

- Owned end-to-end design of Fuse X1, air handling, and co-developed optical subsystems, delivering production CAD in Onshape and driving PLM/DFMA to ensure 95%+ part yields.
- Designed, built, coded, and validated a 50+ part laser purge characterization jig (\$10K+ BOM Cost); integrated mechanical and electrical systems, and managed the deployment of the test hardware for overseas manufacturing.
- Developed and fabricated a cost-effective (<\$500) compression molding press, leveraging additively manufactured tooling and a closed-loop PID thermal control system to successfully manufacture carbon fiber composites.
- Optimized material selection for optical purges by simulating high-thermal environments with porous media models, achieving strong correlation between Ansys Fluent data and physical in-printer performance.
- Rapidly cycled prototype iterations, assembling mechanical and electrical components, triaging and solving issues.

Tesla

Robotics Hardware Engineer - Optimus

May - Aug. 2025

Palo Alto, CA

- Drove calibration repeatability and accuracy validation by developing MOCAP metrology study procedures (DOE), executing tests, and performing statistical analysis with Minitab to reduce assembly time by 15 minutes.
- Analyzed mainline operations using PFMEA to improve operator ergonomics and balance station workloads, reducing cycle time and enabling a transition to mass production with a significant increase in unit output.
- Engineered novel processes for installing robotic limbs and components, ensuring safety compliance, optimizing design for assembly (DFA), and enabling additional tasks during position-controlled or autonomous operation.
- Improved bot accuracy and kinematic understanding by designing dimensional measurement strategies and conducting a first-principles analysis of end-effector error sources, identifying key factors to better controllability.
- Designed poka-yoke jigs and fixtures in CATIA V6 to ensure repeatable and error-proof assembly processes.
- Triageed and resolved issues with firmware and hardware for robots in production to maintain build plan targets.

SKILLS

- **CAD & Analysis:** SOLIDWORKS, CATIA (V5/V6), ENOVIA, Onshape, AutoCAD, Ansys FEA & Fluent.
- **Manufacturing:** 3D Printing (SLA/SLS/FDM), Sheet Metal, GD&T, Tolerance Stackup, Statistical Process Control.
- **Design:** First Principles Analysis, Iterative Design, DFMA, DFMEA, Poka-Yoke, Design Verification & Validation.
- **Software:** Python, C++, ROS2, MoveIt2, PyTorch, Imitation/Reinforcement Learning, VLA, MATLAB.